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Communication

Vinylogous Anionic Ring-Opening Polymerization of Vinylidenecyclopropanes with Native C−H Bonds as the Dormant Species

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Cite This: *J. Am. Chem. Soc.* 2025, 147, 7196−7202
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ABSTRACT: Since its inception in 2023, the concept of "C−H bonds as the dormant species" to control chain growth has emerged as a promising, though still evolving, strategy for precision polymer synthesis. Beyond offering a simpler polymerization method, this concept is expected to promote unprecedented reaction pathways. In this study, we report the first vinylogous anionic ring-opening polymerization (ROP) of vinylidenecyclopropanes. Cooperative La(OTf)$_3$/DBU catalysis, in conjunction with various sp^3 and sp^2 C−H bonds, has streamlined the synthesis of well-defined alkyne polymers, which are amenable to further chemical modifications. Control experiments and density functional theory (DFT) calculations provide insights into the origin of selectivity and the role of dormant C−H bonds in the reversible-deactivation equilibrium.

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Over the past few decades, controlled and living polymerizations (CLPs) of various mechanisms have significantly advanced polymer synthesis and materials science.[1−8] These techniques enable the preparation of well-defined polymers with precisely controlled molecular weight (MW), low dispersity (*Đ*), and tailored structures and topologies. A key principle underlying these achievements is the reversible interconversion between dormant chain-end groups and the active propagating species, namely reversible deactivation,[3−6,9] which minimizes termination reactions to secure high-level control over polymer chain growth. Notably, conventional CLPs typically rely on heteroatom-based dormant species, which are relatively unstable and often lead to polymeric materials with reduced thermal stability.[10−15]

An emerging strategy for reversible deactivation involves the use of native C−H bonds, providing an unprecedented solution to the heteroatom issue. In 2023, our group introduced the concept of "C−H bond as the dormant species" (Figure 1a),[16] and, for the first time, demonstrated its proof of concept in the anionic ring-opening polymerization (ROP) of cyclopropanes (Figure 1b).[17,18] The key to success lies in the catalytic synergy between a Lewis acid and an organic base, which not only activates a wealth of sp^3 C−H bonds but also mediates reversible deactivation via proton transfer. Furthermore, this strategy tamed the challenging S_N1-type zwitterionic ROP of cyclobutanes,[19] transforming a step-growth polymerization into a controlled chain polymerization (Figure 1b).[20] However, these methods have thus far been limited to local reactivity within the ring, yielding only C3 and C4 polymers. We envisioned extending this new concept to harness global reactivity, where the incorporation of pendent functionalities for main-chain bond formation is realized, ultimately producing polymeric materials with advanced structures and properties.

The principle of vinylogy,[21] transmitting electronic effect to a distal position via interposed conjugation, is one of the most extensively studied approaches to achieving global reactivity. Despite π-like molecular orbitals in cyclopropanes,[22] their vinylogous ring-opening transformations via ionic pathways remain underexplored, with few exceptions observed on vinylcyclopropanes.[23−25] In contrast, vinylidenecyclopropane (VDCP),[26] a versatile building block featuring the strained cyclopropane and an allene moiety, has not been explored for ionic polymerizations. Shi et al. demonstrated that under Lewis acid catalysis, sulfonamides appended to VDCPs can add to the C_5-position,[27] whereas intermolecular reactions proceeded via the C_2-pathway (Figure 1c).[28] These examples inspired us to explore general carbon nucleophiles for explicit C_5-addition, thereby achieving vinylogous reactivity. If successful, it could lead to propagation with excess VDCPs, building alkynes in each repeating unit. Here, we report the first vinylogous anionic ROP of VDCPs,[29] utilizing the Lewis acid/organic base/C−H bond triple system previously developed in our laboratory.[16] The oxophilic lanthanide triflate coordinates to the 1,3-dicarbonyl moiety in VDCP, inducing charge relocation through π-conjugation, which enhances the C_5-electrophilicity. Concurrently, the base activates the C−H bond to generate an anion for the propagation (Figure 1d). This synchronous activation eventually enables rapid and controlled ROP of VDCPs with exclusive vinylogous selectivity.

Received: December 16, 2024
Revised: February 17, 2025
Accepted: February 19, 2025
Published: February 24, 2025



ACS Publications
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https://doi.org/10.1021/jacs.4c17911
J. Am. Chem. Soc. 2025, 147, 7196−7202

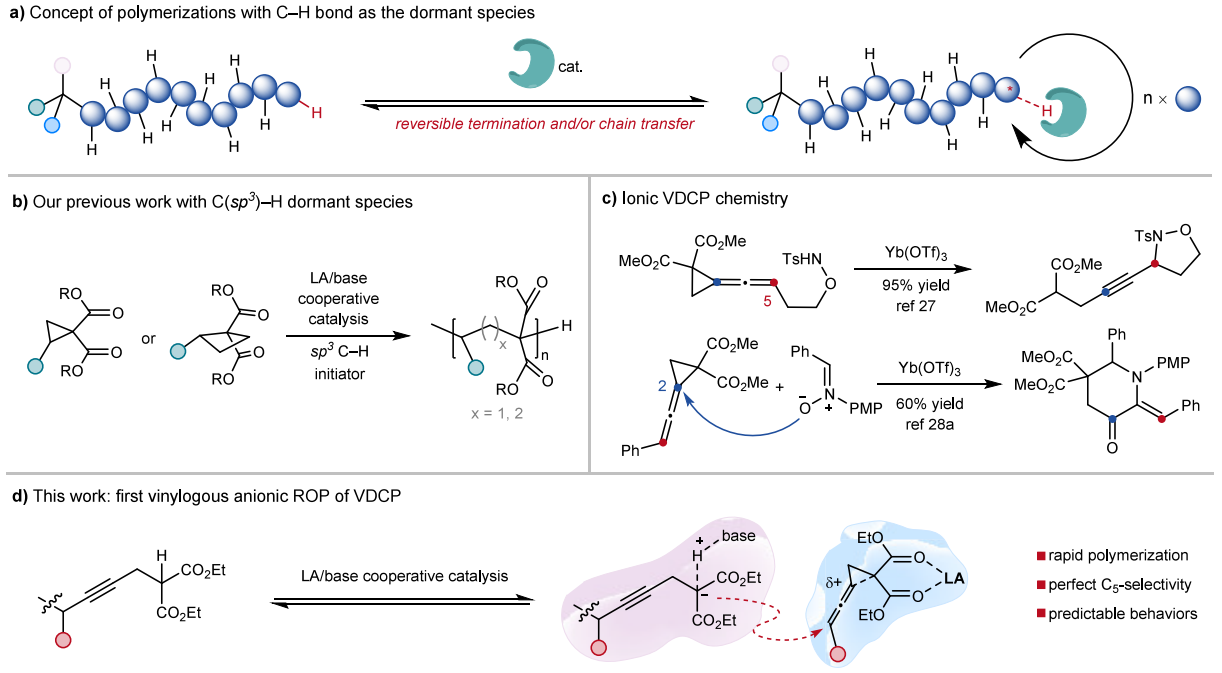
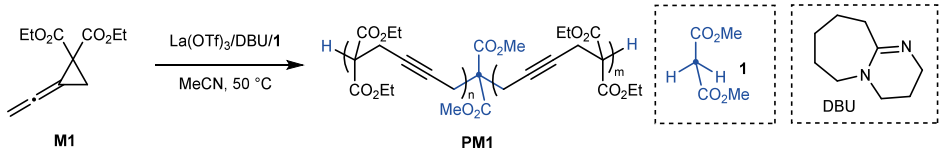


Figure 1. Background and inspiration for developing vinylogous anionic ROP of VDCPs.

Table 1. Polymerization Development[a]



entry	**M1**/La(OTf)$_3$/DBU/**1**	convers (%)[b]	$M_{n,theo}$ (kDa)[c]	M_n (kDa)[d]	$Đ$[d]
1	100/5/5/5	99	4.3	4.3	1.07
2	100/0/5/5	<5			
3	100/5/0/5	<5			
4	100/5/5/0	99		18.3	1.10
5[e]	100/5/5/5	99	4.3	4.3	1.06
6[f]	100/5/5/5	99	4.3	5.0	1.06
7	100/5/2/5	98	4.2	4.4	1.06
8	100/2/2/5	99	4.3	5.0	1.06
9	100/2/1/5	87	3.8	3.8	1.12
10[g]	100/1/2/5	99	4.3	4.4	1.04
11[g]	100/1/1/5	67	2.9	3.7	1.07
12	100/2/2/2	99	10.6	10.2	1.05
13[g]	100/1/1/1	99	21.1	20.0	1.06
14[g]	100/0.5/0.5/0.5	95	40.0	32.0	1.07
15[g]	100/1/1/0.5	99	42.1	44.9	1.08
16[g]	100/0.5/1/0.25	99	84.1	79.4	1.11
17[g]	100/1/1/0.1	99	210.1	115.7	1.22

[a]0.2 mmol scale in 0.6 mL of anhydrous MeCN at 50 °C. [b]Measured by scrude ^{1}H NMR. [c]$M_{n,theo}$ = MW(initiator) + MW(**M1**) × conversion × ([**M1**]/[initiator]). [d]Determined by GPC (with polystyrene calibration). [e]With 2.5 mol % of H_2O. [f]With 20 mol % of *t*-BuOH. [g]0.6 mL of 1:1 MeCN:EtOAc. OTf = triflate. GPC = gel permeation chromatography.

We began our investigations by polymerizing diethyl 2-vinylidenecyclopropane-1,1-dicarboxylate (**M1**) with 5 mol % of 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) as the base and dimethylmalonate (**1**) as the initiator, in MeCN at 50 °C to assess the impact of Lewis acid catalysts on the polymerization (Supporting Information Table S1). The use of 5 mol % La(OTf)$_3$ achieved 99% monomer conversion within 10 min, producing **PM1** with a M_n of 4.3 kDa being close to its theoretical value and a low $Đ$ of 1.07 (Table 1, entry 1). No allenyl groups were detected in **PM1**, indicating the polymerization proceeded via the vinylogous pathway. Control experiments showed that neither La(OTf)$_3$ nor DBU was capable of driving the polymerization (entries 2 and 3); in the absence of **1**, DBU initiated a vinylogous zwitterionic but uncontrolled polymerization (entry 4).[16] Notably, this protocol was tolerant to small amounts of protic additives,

https://doi.org/10.1021/jacs.4c17911
J. Am. Chem. Soc. 2025, 147, 7196−7202

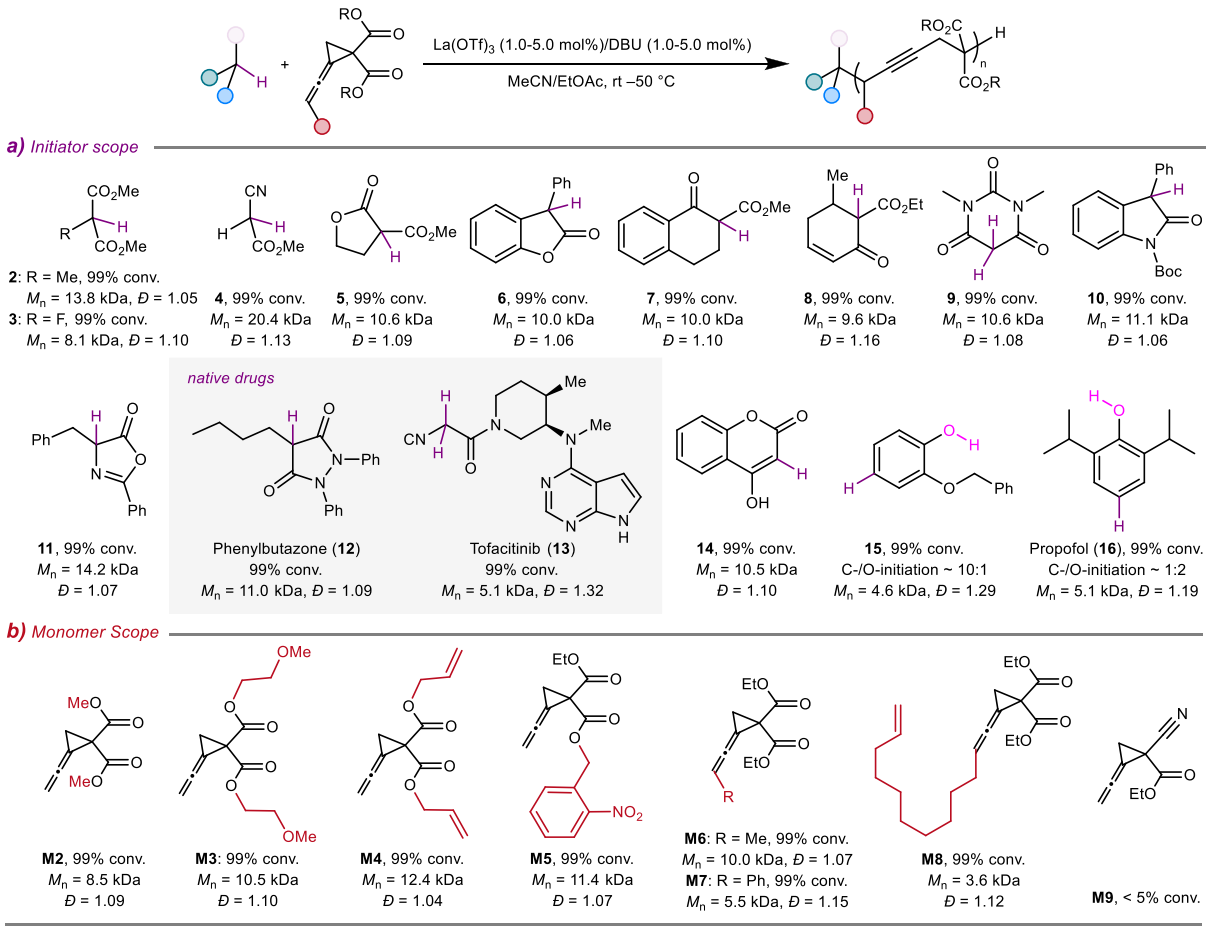


Figure 2. Scope of initiators and monomers.

such as H_2O and *t*-BuOH (entries 5 and 6), likely due to the reversible termination mechanism. Decreasing the amounts of DBU and/or $La(OTf)_3$ had little effect on the polymerization outcomes, which consistently yielded **PM1** with predictable M_n of 3.7−5.0 kDa and low $Đ$ of 1.04−1.12 (entries 7−11), supporting the reversible chain transfer mechanism via protonation. Importantly, increasing the feed ratios of **M1** and **1** (50−400) resulted in polymers with M_n ranging from 10.2 to 79.4 kDa, while maintaining low dispersities of 1.05−1.11 (entries 12−16). It is noteworthy that, due to the superior solubility of $La(OTf)_3$ in EtOAc, the use of mixed MeCN/EtOAc solvent improved the control over polymer chain growth in polymerizations of [**M1**]/[**1**] ≥ 100. However, when targeting a higher degree of polymerization (DP = 1000), the M_n (115.7 kDa) deviated significantly from its theoretical value, likely due to the occurrence of competing DBU-initiated zwitterionic polymerization.

We next explored the generality of C−H bonds (Figure 2a). A wide range of compounds, including dimethylmalonates (**2**−**3**), α-substituted lactones (**5**−**6**), and β-ketoesters (**7**−**8**), proved to be excellent initiators/chain-transfer agents, all resulting in high monomer conversions (99%) and yielding α-functionalized **PM1** with ω-C−H chain ends (M_n = 8.1−13.8 kDa, $Đ$ = 1.05−1.16). Polymerization with methyl cyanoacetate (**4**) led to **PM1** of 20.4 kDa, which is twice as high as its theoretical value. This is likely due to the slow initiation from **4**, as it typically does not form bidentate coordination with the Lewis acids. Complex heterocyclic compounds, such as 1,3-dimethylbarbituric acid (**9**), oxindole (**10**), azlactone (**11**), and several native drugs, including phenylbutazone (**12**) for treating ankylosing spondylitis and tofacitinib (**13**) for rheumatoid arthritis, were also effective in initiating and controlling the polymerization of **M1**.

Interestingly, polymerization with 4-hydroxycoumarin (**14**) also proceeded well to yield **PM1** of 10.5 kDa and $Đ$ = 1.10. This is the first example of using acidic sp^2 C−H bonds as the dormant species. Encouraged by this result, we began to explore nonacidic sp^2 C−H bonds. We hypothesized that deprotonation of the phenolic O−H bond (pK_a ∼ 18.0 in DMSO[30]) by DBU (pK_a 24.3 in MeCN[31]) could enhance its ortho- and para-nucleophilicity. In practice, stoichiometric reaction of 2-benzyloxyphenol (**15**) with **M1** under conditions of $La(OTf)_3$ (10 mmol %) and DBU (10 mmol %) yielded para-propargylated product in 45% isolated yield (Figure S65), supporting our hypothesis. Interestingly, the polymerization of **M1** with **15** featured dual C/O initiations (Figure S67). Propofol (**16**) exhibited significant O reactivity in its stoichiometric reaction with **M1**, and the C/O selectivity was ∼1.2:1 (Figures S68 and S70); while for polymerization, dual C/O initiations and single O initiation were observed with a ratio of ∼1.4:1 (Figure S72). These findings highlight the versatility of cooperative $La(OTf)_3$/DBU catalysis in promoting sp^2 C−H bond initiation.

The scope of VDCPs was further explored (Figure 2b). Polymerizations of dimethyl 2-vinylidene-1,1-dicarboxylate (**M2**), di(2-methoxyethyl) 2-vinylidene-1,1-dicarboxylate (**M3**), and diallyl 2-vinylidene-1,1-dicarboxylate (**M4**), under conditions of [M]/[$La(OTf)_3$]/[DBU]/[**1**] = 100/2/2/2,

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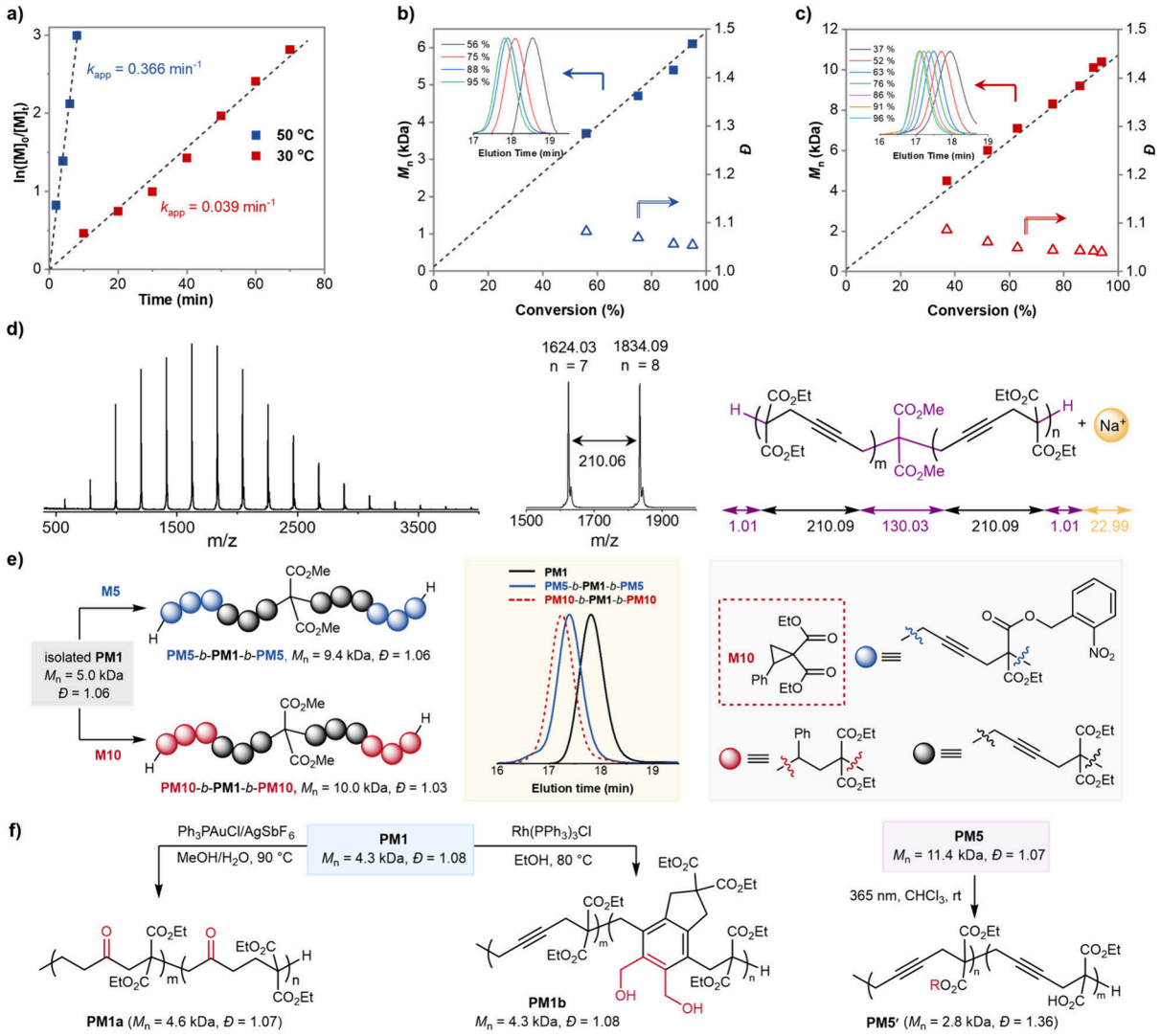


Figure 3. (a) Polymerization kinetics. Plots of M_n and Đ as a function of monomer conversion at (b) 50 and (c) 30 °C. (d) MALDI-TOF analysis of **PM1**. (e) Chain-extension experiments. (f) Postpolymerization modifications.

successfully yielded **PM2**−**PM4** with target M_n of 8.5−12.4 kDa and low Đ of 1.04−1.10. A photocleavable *o*-nitrobenzyl ester group in the monomer (**M5**) was tolerated, leading to degradable **PM5** (11.4 kDa, Đ = 1.07). Excitingly, monomers bearing a C_5-substituent, including methyl (**M6**), phenyl (**M7**), and 10-ene-undecyl (**M8**) groups, proved polymerizable. The M_ns of obtained **PM6** (10.0 kDa) and **PM7** (5.5 kDa) were close to their theoretical values (11.2 kDa and 5.8 kDa, respectively); while **PM8** exhibited a lower M_n of 3.6 kDa ($M_{n,theo}$ = 7.1 kDa), probably due to background polymerization yielding cyclic polymers, as indicated by the matrix-assisted laser desorption/ionization time-of-flight mass (MALDI-TOF MS) spectrometry (Figure S44). These results addressed the limitation on terminal substitutions in the radical polymerization.[32] Replacing one of the 1,1-dicarboxylates with a cyano group (**M9**) led to <5% monomer conversion, likely due to the loss of bidentate coordination to La(OTf)$_3$.

To evaluate the control imparted by dormant C−H bonds, several experiments were performed. First, progress analysis revealed temperature-dependent first-order polymerization kinetics, with apparent rate constants (k_{app}) of ∼0.366 min^{-1} at ∼50 °C and ∼0.039 min^{-1} at ∼30 °C (Figure 3a). A linear increase in the M_n with monomer conversion, as well as consistently low dispersities through the entire course of polymerization, were observed (Figure 3b,c). Second, MALDI-TOF mass spectrum of **PM1** showed one set of molecular peaks corresponding to a repeat unit mass of 210.06 Da and a dimethylmalonate chain-end group mass (plus Na^+) of 155.03 Da (Figure 3d). Third, using an isolated **PM1** (5.0 kDa, Đ = 1.06) as the macroinitiator, chain-extension experiments with **M5** and diethyl 2-phenyl-1,1-dicarboxylate (**M10**)[16] produced ABA-type block copolymers, **PM5**-*b*-**PM1**-*b*-**PM5** (9.4 kDa, Đ = 1.06) and **PM10**-*b*-**PM1**-*b*-**PM10** (10.0 kDa, Đ = 1.03), as confirmed by NMR characterization (Figures S73 and S74) and shifts in the retention time of GPC traces (Figure 3e).

Since postpolymerization modification (PPM) has become a powerful strategy for creating novel functional polymers, we were interested in expanding the chemical space of poly(VDCPs). As shown in Figure 3f, a gold-catalyzed hydration reaction[26] quantitatively converted **PM1** (M_n = 4.3 kDa, Đ = 1.08) into polyketones (**PM1a**, M_n = 4.6 kDa, Đ = 1.07); while a rhodium-catalyzed [2 + 2+2] cycloaddition reaction[33] with 2-butyne-1,4-diol built ∼50% in-chain benzene motif, producing **PM1b** with M_n = 4.6 kDa and Đ = 1.07. Finally, UV light (365 nm)-induced photolysis[34] removed ∼65% of preinstalled *o*-nitrobenzyl ester groups in **PM5** (11.4 kDa, Đ = 1.07) within 8

https://doi.org/10.1021/jacs.4c17911
J. Am. Chem. Soc. 2025, 147, 7196−7202

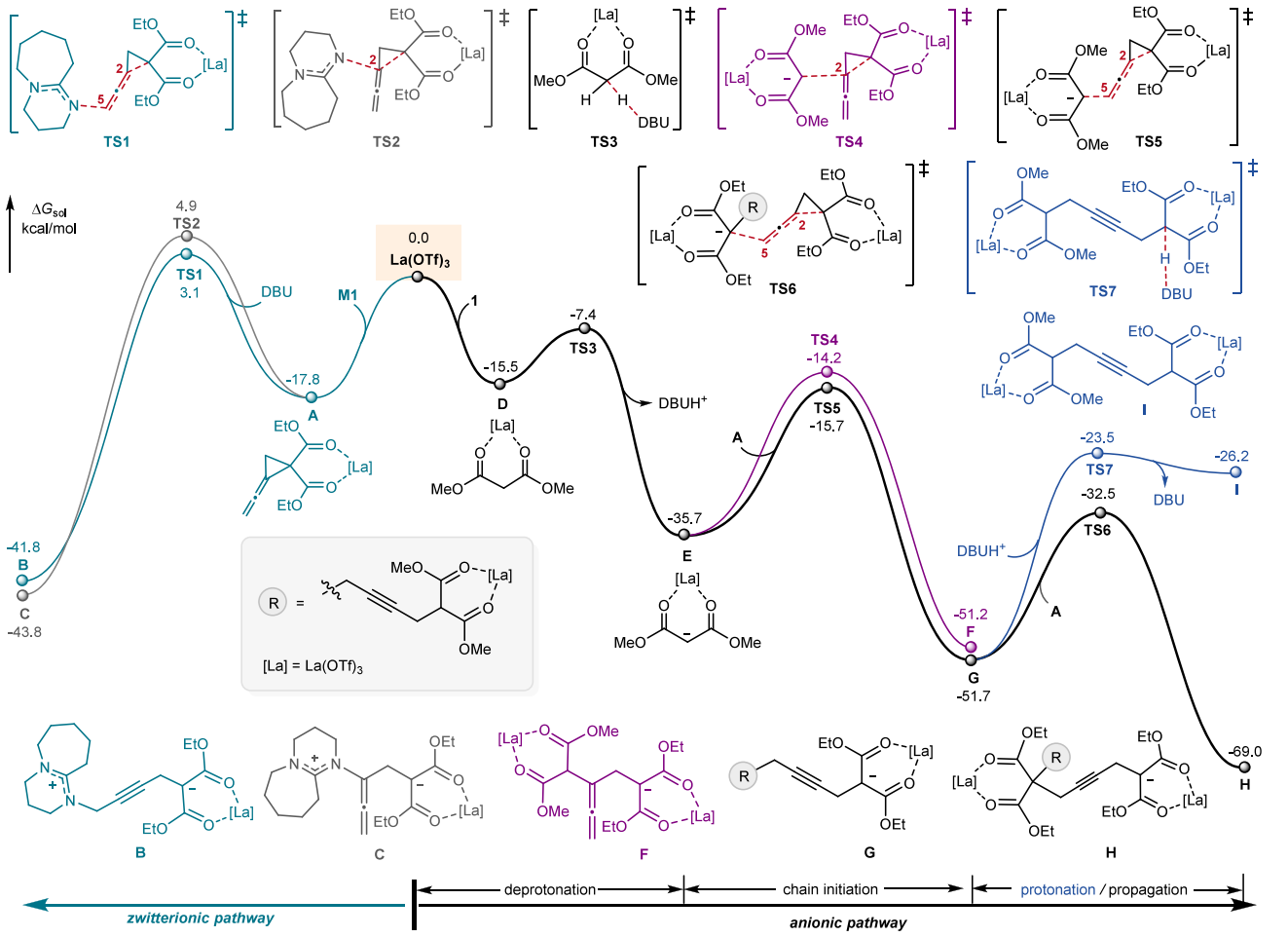


Figure 4. Computational investigations of polymerization mechanism.

h, accompanied by unexpected degradation to oligomers (M_n = 2.8 kDa, $Đ$ = 1.36), probably via radical chain scission pathways.

To gain deeper insights into the ROP of **M1**, we performed density functional theory (DFT) calculations, in which both the DBU-initiated zwitterionic pathway and the anionic pathway were considered. In general, La(OTf)$_3$ preferentially coordinates to 1,3-dicarbonyls, giving energetically stable species **A** (by 17.8 kcal/mol) and **D** (by 15.5 kcal/mol). In the DBU-initiated zwitterionic pathway (Figure 4, left), although both reactions are significantly exothermic, the nucleophilic addition of DBU to **A** at the C_5-position, via the transition state **TS1**, is 1.8 kcal/mol kinetically more favorable than the addition at inner C_2-position, via the transition state **TS2**. In the anionic pathway (Figure 4, right), we found that the deprotonation of **D** by DBU through **TS3** ($\Delta G^{‡}$ = 8.1 kcal/mol) is much easier than DBU initiations ($\Delta G^{‡}$ = 20.9 and 22.7 kcal/mol), thereby outcompeting the zwitterionic pathway.

Starting from the anionic species **E**, both local (C_2-selective) and vinylogous (C_5-selective) ring-opening reactions are thermodynamically favorable. However, the latter process via **TS5** is kinetically more favorable, by 1.5 kcal/mol, than the former via **TS4**. These findings align with our experimental observations, which suggest a predominant vinylogous anionic ROP mechanism. Due to the release of ring strain energy, further propagation from the propargyl carbon anion **G** with **A** is thermodynamically favorable, with an energy barrier of 19.2 kcal/mol. By contrast, the protonation reaction of the anion **G** with DBU-H$^+$ has to overcome an energy barrier of 28.2 kcal/mol. Notwithstanding, the tolerance to specific amounts of protic additives in the polymerization (Table 1, entries 5 and 6) supports the viability of reversible termination. Arguably, barrierless chain transfer events via proton exchange likely dominate the reversible deactivation equilibrium (Table 1, entries 7−11).

In summary, we have developed a La(OTf)$_3$/DBU/C−H triple system for the first vinylogous anionic ROP of vinylidenecyclopropane monomers, producing diverse main-chain alkyne polymers with tailored molecular weights and low dispersity. A number of sp^3 and sp^2 C−H bonds qualify in initiating the polymerization and regulating the chain growth via reversible termination and chain transfer. This strategy enables unprecedented polymer chemistry of cyclopropanes and is expected to transform precision polymer synthesis and related materials science.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c17911.

Experimental and computational details; characterization data for all monomers and polymers; ^{1}H NMR and ^{13}C NMR spectra; and additional relevant references (PDF)

https://doi.org/10.1021/jacs.4c17911
J. Am. Chem. Soc. 2025, 147, 7196−7202

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Key Research and Development Program of China (2024YFA1509202), the National Natural Science Foundation of China (Nos. 22471258, 22271001, and 52425306), and the CAS Project for Young Scientists in Basic Research (YSBR-094). This work was partially carried out at the Instruments Center for Physical Science, University of Science and Technology of China (USTC). Y.L. and G.L. acknowledge Hefei Advanced Computing Center and the High-Performance Computing Platform of Anhui University for computational support.

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https://doi.org/10.1021/jacs.4c17911
J. Am. Chem. Soc. 2025, 147, 7196−7202